|METRIC|
DOD-P-28682(YD)
5 November 1987

MILITARY SPECIFICATION

CONCRETE PLANT, MOBILE, SEMITRAILER MOUNTED, 6 CUBIC METERS BULK MATERIAL CAPACITY (METRIC)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a complete mobile plant consisting of a combination concrete mixing unit and material transporter, separate diesel engine powered, capable of producing 6 cubic meters (m^3) of concrete at the job site.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservations, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair parts, Accessories, and Kits, Mechanical, Packaging of.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 3895

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MIL-E-10062 - Engines: Preparation for Shipment and Storage.
MIL-T-22085 - Tapes, Adhesive, Preservation and Sealing.
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-P-46093 - Primer Coating, Synthetic, (For Brake Drums).

FEDERAL

UU-T-81 - TAG, Shipping and Stock
QQ-S-781 - Strapping, Steel, and Seals.
TT-P-664 - Primer Coating, Synthetic, Rust Inhibiting, Lacquer-Resisting.
PPP-T-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MS75021 - Connector, Receptacle, Electrical 12-Contact, Intervehicular, 28 Volt, Waterproof.

2.1.2 Other Government documents, and publications. The following other Government documents, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C685 - Concrete Made by Volumetric Batching and Continuous Mixing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. The plant shall consist of a tandem axle semitrailer with mounted concrete materials proportioning, mixing, and delivery system. System shall include sand, stone, and cement bins, mix conveyor, mix proportioning system, main drive train, vibrators, compressed air system, and operating controls.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 6.1.1).

3.3 Standard commercial product. The equipment shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the equipment being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Safety. All rotating or reciprocating parts that are of such a nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The plant shall comply with applicable OSHA Regulations.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using

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materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Construction. Construction of the equipment shall be in accordance with the best engineering practice. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to the equipment.

3.8 Capacity. The plant shall be capable of transporting sufficient unmixed material (bulk cement, sand, crushed stone or gravel, and water) to the job site to produce not less than 6 m^3 of concrete based on design formulas requiring as much as six bags of cement, 0.4 m^3 of fine aggregate and 0.6 m^3 of coarse aggregate (not exceeding 38 millimeter (mm) size) per cubic meter of mixed concrete. Bins and water tank shall have the following minimum capacities:

Sand bin	- 3.7 m^3 dry sand.
Stone bin	- 5.3 m^3 dry stone.
Cement bin	- 1.8 m^3
Water tank(s)	- 1515 liters.

3.9 Performance. The plant shall function as a complete delivery and production system capable of producing various design mix concrete from zero slump to very wet mixes. The plant shall be capable of continuous or intermittent mixing and delivery as required and shall be capable of operating and producing concrete in accordance with ASTM C685. The unit shall be capable of being towed at speeds as great as 16 kilometers per hour (kmh) over unimproved roads and at speeds as great as 88 kmh over improved roads.

3.9.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree (°) angle to the coupled towing vehicle without cramping, sideslipping, or damage to the semitrailer or the towing vehicle.

3.9.2 Tracking ability. The semitrailer shall conform to the tracking requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(e).

3.9.3 Brake performance. Service and parking brakes shall comply with the performance requirements of DoT Federal Motor Vehicle Safety Regulation section 393.52.

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3.9.4 Ratings. Trailer ratings shall be the manufacturer's published ratings. Component and trailer ratings shall not be raised to meet the requirements of this specification.

3.10 Semitrailer unit. The plant shall be mounted on a drop frame type semitrailer designed to withstand shock loads incumbent with performance requirements specified herein. The semitrailer unit shall be equipped with full air service brakes controlled from the towing vehicle. Service brakes, lighting reflectors, and associated equiment shall conform to DoT Federal Motor Carrier Safety Regulations. The upper fifth wheel plate shall be designed for coupling to a 1371 mm ± 26 mm in height, full oscillating, and fore and aft rocking type fifth wheel 914 mm in diameter and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.70(b), (e), and (h). The kingpin shall conform to SAE J700. Support jacks shall be provided in sufficient number and location to support the semitrailer unit in its operating position when the semitrailer is unhooked from the towing vehicle. Axle and suspension ratings shall be at least equal to the imposed loads.

3.10.1 Wheels, rims, tires, and tubes.

3.10.1.1 Wheels, rims, and tires. The semitrailer shall be equipped with dual wheels, when specified (see 6.1.1), wide base wheel shall be on each axle. Tires shall be tube or tubeless type with highway tread. The rims and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels of the semitrailer. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with semitrailer loaded with rated payload. Tires shall be of not less than 100 level quality and shall be of domestic manufacture.

3.10.1.2 Tire carrier. One tire carrier shall be provided and installed in a readily accessible location under the chassis. Means shall be provided for securing the tire within the carrier to prevent accidental loss.

3.10.1.3 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy-duty type, and shall be of proper size for tires furnished.

3.10.1.4 Spare wheel. When specified (see 6.1.1), a spare wheel or rim shall be mounted on the tire carrier.

3.10.1.5 Spare tires. When specified (see 6.1.1), a spare tire mounted on wheel shall be provided.

3.10.2 Brakes.

3.10.2.1 Service brakes. Service brakes shall be of the full air type, and shall conform to DoT Motor Carrier Safety Regulations, sections 393.45 through 393.47, and as specified herein. The braking system shall include breakaway features, manual or automatic slack adjusters, piping, hose connections, gladhands, spring loaded dustcovers or dummy gladhands equipped with security chains, and all other components required for a complete air-brake system. The braking system shall be installed in a manner which

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provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims. Gladhands shall conform to SAE J318.

3.10.2.2 Parking brake. The semitrailer shall be equipped with a parking brake. The parking brakes shall be spring applied or air-diaphragm type. The brakes shall hold the semitrailer with rated payload on a 10 percent grade despite the depletion of the compressed air supply. The brakes shall be automatically applied upon discontinuation of the emergency air line and under emergency conditions.

3.10.3 Electrical system.

3.10.3.1 Lighting. The electrical lighting system shall be 12 volt (V) potential. The lighting system shall conform to DoT Motor Carrier Safety Regulations, secitons 393.14, 393.20, 393.22, 393.25 through 393.29, 393.32, and 393.33. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails. Clearance lights shall have replaceable lamps with twist or snap-on lenses. Turn signal lamps shall conform to SAE J588. The front of the semitrailer shall be equipped with a 12V, 7 contact receptacle conforming to SAE J560, with the receptacle located and the conductors connected and color coded as specified therein.

3.10.3.2 Independent 24V direct current (DC) electric system. When specified (see 6.1.1), a 24VDC lighting system, independent of the 12VDC lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24VDC system. The 24VDC system shall also conform to the DoT Federal Motor Carrier Safety Regulations.

3.10.3.3 Receptacle for 24VDC electric system. When an independent 24VDC system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS75021, part number MS75021-1. The receptacle shall be located in accordance with SAE J702. The 24VDC, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow).
Contact D. Connect to ground (white).
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown).
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24VDC system, the normal 12VDC turn signal lights shall function both as turn signals and stop lights, and the normal 12VDC stop lights will not be operational when the semitrailer is connected to a towing vehicle with as 24VDC power supply. Because of this condition, the stop light (red) circuit is not connected to the 24VDC, 12-contact receptacle.

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3.11 Rear wheel splash and stone throw protection. Rear wheels shall have mud flaps at rear. Splash and stone throw protection shall be in accordance with SAE J682.

3.12 Toolbox. A toolbox shall be provided. The toolbox shall be large enough to store all tools required for field service or maintenance, but shall not have external closed dimensions less than 350 mm in length, 150 mm in width, and 150 mm in height. The toolbox shall be of nominal 1.88 mm (US revised standard gage No. 14) thick steel, with a hinged lid and a trunk drawbolt to keep the lid secure when vibrated. The toolbox shall be mounted in a protected accessible location.

3.13 Plant components.

3.13.1 Sand and stone bins. The sand and stone bins shall be mounted on the semitrailer chassis and shall be of all welded steel construction, braced, and gusseted for adequate strength and rigidity. The bin side plates and divider plates shall be designed to prevent bridging and clogging of aggregates. Each bin shall have a wide top opening for loading by front end tractor loader, conveyor, or other means generally used for loading open top bins or truck bodies with dry granular material. A control gate, or other suitable means shall be provided to regulate the flow of aggregate. The gate shall be provided with a position locking device to lock the gate in the desired position. A dial gage, divided into numbered increments, shall be provided on each gate to indicate position of gate opening for regulating material flow for the required mix. The sand bin shall be provided with not less than two pneumatic vibrators on the outside of the bin to insure against bridging and provide continuous, uniform flow of the aggregate.

3.13.2 Cement bin. The cement bin shall be mounted on the semitrailer and shall be of similar construction as the stone and sand bin. The bin shall have a closed top with a gasketed weathertight loading door located in the top center of the bin. The door opening shall be of sufficient size to allow the bin to be loaded from cement bags or bulk cement by conveyor, bucket elevator, auger, overhead bin, or any other method generally used for the loading of dry cement. The bin shall be equipped with not less than two external pneumatic vibrators to insure against bridging of cement and assure continuous uniform flow of cement. A cement meter register, with totalizer, shall be provided to indicate the amount of cement introduced into a mix at any time during production and to record total cement discharge of all deliveries.

3.13.3 Mixing system. The mixing system shall consist of a mix conveyor auger of welded steel construction with replaceable wear plates, and sand stone feeder conveyor with multiple design cover construction for abrasion resistance, and manufacturers standard cement feeder auger and system. Conveyor and feeder shall be mechanically or hydraulically driven from the plant drive engine. A clutch shall be installed on the main line shaft to control motive operation of the sand, stone, and cement conveyors metering system. The cement bin clutch shall enable discharge of sand and stone bins without discharge of cement and shall disengage the meter register.

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3.13.4 Delivery chutes. Two delivery chutes shall be provided to give a total delivery distance of 3.6 meters including the mix conveyor. Delivery chutes shall provide not less than 170° horizontal movement.

3.13.5 Water system. A pressure water system shall be provided and shall include water tank(s), water level gage, pump, flowmeter, flow control valve, strainers, washout hose, and drainout valves. The water tanks(s) interior surfaces shall be coated with Bitumastic or equal material to prevent corroison. The level gage, flowmeter, and flow control valve shall be calibrated in liters per minute. Strainers shall be provided to remove extraneous materials from the water and to protect the pump and metering devices. Adequate drain valves shall be provided to assure complete drainage of the water system for protection in subfreezing weather. When specified (see 6.1.1), an additive mixture system shall be provided. The system shall proportion chemical additives into the concrete mix.

3.13.6 Air system. The plant air system of not less than .35m^3 per minute air shall be provided (see 3.13.8). The air system shall consist of heavy-duty air hoses, filter, lubricator, air receiver and all necessary valves.

3.13.7 Controls. Operating controls shall be located at or near the left rear corner of the unit and shall be wihin easy reach of the of the operator while standing on the ground. All controls shall be clearly identified for their intended function with permanently affixed identification plates made of corrosion-resistant material.

3.13.8 Engine. The engine furnished for the plant shall be a diesel engine having horsepower, torque, and speed characteristics to satisfactorily meet all the requirements specified herein. The engine shall comply with all applicable EPA regulations. The engine shall be furnished complete with accessories as follows:

a. Fuel tank with sufficient capacity for 8 hours operation.
b. A 12V electric cranking system.
c. A generator or alternator with a rating of not less than 17 amperes.
d. Governor.
e. Air cleaner with precleaner.
f. Control panel with starter switch, ammeter, oil pressure gage, cooling liquid temperature gage (for liquid-cooled engines only), and hour meter.
g. Air compressor pump.

3.14 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and

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treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than .06 mm over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall be No. 14064 green conforming to FED-STD-595.

3.15 Lifting and tiedown attachments. The unit shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the unit on the carrier when shipped.

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.17 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates. Nomenclature shall be "CONCRETE PLANT, MOBILE, TRAILER MTD, DED 6CU M CAP, COMBINATION MATERIALS SYSTEM."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one concrete plant when a first article is required (see 3.2 and 6.1.1). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets

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the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining plants to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each plant shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. When first article is required, the first article shall be subject to the tests specified in 4.4.1 through 4.4.5. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all plants until corrective action has been taken.

4.4.1 Vehicle weight. The plant shall be loaded to the capacities specified in 3.8 and weighed to determine weight and distribution of weight on the fifth wheel and axles. The imposed loading of the fifth wheel and axles shall be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet contract requirements, and to determine conformance to 3.8 and 3.10.

4.4.2 Road test. Road test shall consist of coupling the semitrailer unit loaded with rated payload to a truck-tractor. The tractor combination shall be driven a distance of not less than 32 km under the conditions specified in 3.9. At least 20 percent of the distance shall be over hard uneven terrain. Not less than five sudden stops shall be made from a speed of not less than 20 kmh. Turning ability specified in 3.9.1 and tracking ability specified in 3.9.2 shall be observed to ascertain conformance to the specification. After the test, the unit shall be examined for evidence of misalinement, binding, or other malfunction.

4.4.3 Landing support test. Landing support shall be tested to verify conformance to 3.10.

4.4.4 Operation test. With the plant loaded to the capacities specified in 3.8 and the plant air system connected to a truck-tractor air supply, operate the plant to mix a total of 6 cubic meters of concrete in one cubic meter batches, changing the aggregate proportions to simulate different mix designs. Observe the operation and performance of all plant components specified in 3.13 and to ascertain conformance to the specification.

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4.4.5 Lifting and tying down attachment tests. The lifting and tying down attachments shall be tested to verify conformance to 3.15.

4.5 Production tests. The contractor testing system shall, as a minimum, assure that the plant conforms to the physical and dimensional requirements specified and is capable of meeting the performance requirements specified herein.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.1.1).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified UU-T-81 with weatherproof tags conforming to UU-T-81, attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape conforming to PPP-T-60.

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5.1.1.6 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A UU-T-81 weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.7 Gears.

5.1.1.7.1 Exposed gears. All unpainted surfaces of exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.7.2 Enclosed gears. Enclosed gears shall be filled to the operating level with the approved lubricant required for operation. Gear mechanism shall be operated to insure coating of all interior surfaces with preservative. The gear housing shall be identified with a UU-T-81 weatherproof tag to indicate, "This housing is filled to the operating level with lubricant required for operation. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.8 Hydraulic systems. The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches onto the ram cylinders. The wraps shall be secured in place with PPP-T-60 waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.1.9 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.10 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.11 Instruments. The dial glass of instruments not protected by a metal housing shall be covered individually or in groups with a fitted piece of plywood secured with tape conforming to PPP-T-60, type IV or MIL-T-22085, type II.

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5.1.1.12 Vehicle bodies. The vehicle body and chassis shall be preserved as specified in MIL-STD-281 for level A (mobile) requirements.

5.1.1.13 Lubrication. Lubricate all grease fittings with type P-11 preservative to include universal joints, chassis suspension and steering linkage, fifth wheels, propeller shaft, and parking brake cable. Wheel bearings requiring a NLGI No. 2 grade shall be lubricated with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-10.

5.1.1.14 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 pounds above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

5.1.1.15 Radiator fronts and engine hoods. When specified (see 6.1.1), radiator fronts and engine hood openings of uncrated mixers shall be covered or sealed as specified herein. Radiator fronts of uncrated mixers shall be covered with waterproofed barrier conforming to PPP-B-1055, class E-2 or M-1. The barriers shall be secured with tape conforming to PPP-T-60, type IV, or MIL-T-22085, type II. All vents, louvers, and other openings of engine hoods shall be sealed with tape specified herein.

5.1.1.16 Service parts. The preservative application criterial and applicabe methods of preservation of MIL-P-116 shall be used to preserve service parts. When specified (see 6.1.1), the service parts shall be preserved in accordance with level A requirements of MIL-R-196, or when parts are not specifically covered in MIL-R-196, an applicable submethod of preservation of MIL-P-116 shall be used.

5.1.1.17 Engines. Engines, engine components, and accessories shall be preserved in accordance with level A requirements of MIL-E-10062, type I, method I or type II, method I as applicable.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The mixers shall be lubricated for operational services as specified in the operator's manual.

5.2 Packing. The packing shall be level A, B, or commercial as specified (see 6.1.1).

5.2.1 Level A.

5.2.1.1 Crated mixers. Unless otherwise specified (see 5.2.1.2 and 6.1.1), each complete mixer shall be packed in a crate conforming to MIL-C-104, type II, class 1, style a. When specified (see 6.1.1) the crate shall conform to type I. The contents shall be anchored, blocked, and braced of the contents shall be in accordance with the applicable box or crate specification and MIL-STD-1186. Strapping for the boxes or crates shall conform to QQ-S-781, class 1, type I or IV, finish B.

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5.2.1.2 Uncrated (mobile) mixers. When specified (see 6.1.1), each complete mixer shall be prepared for mobile shipment in accordance with applicable carrier rules and regulations. Consolidation containers shall be secured on the mixer in a manner not to increase cubage or interfere with towing or lifting the mixer with slings. Support legs (landing gears) shall be used for packing and storage only. For shipment, the trailer front shall be secured and supported by the use of wood cribbing or blockings, as applicable. Openings to the loader and drum shall be sealed with barrier material conforming to PPP-B-1055, class E-2, N-5, N-1 or L-4; or MIL-B-121, type I, grade A to prevent the entry of water.

5.2.2 Level B.

5.2.2.1 Crated mixers. Unless otherwise specified (see 5.2.2.2 and 6.1.1), each complete mixer shall be packed in a crate conforming to MIL-C-3774, type II, style A. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the applicable box or crate specification and MIL-STD-1186. In addition, for items packed in a crate, contents shall be waterproofed with a full shroud in accordance with the crate specification.

5.2.2.2 Uncrated (mobile) mixers. When specified (see 6.1.1), each complete mixer shall be prepared for mobile shipment as specified for level A.

5.2.3 Commercial. Mixers shall be packed for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval (see 3.2, 4.2.1,and 6.3).
c. When wide base wheel is required (see 3.10.1.1)
d. When spare wheel is required (see 3.10.1.4).
e. When spare tire is required (see 3.10.1.5).
f. When a 24 volt electric system is required (see 3.10.3.2 and 3.10.3.3).
g. When additive mixture sysem is required (see 3.13.5)
h. Level of preservation and packing required (see 5.1 and 5.2).
i. When repair parts are to preserved in accordance with level A requirements of MIL-R-196 (see 5.1.1.16).
j. When type I crate is required (see 5.2.1.1 or 5.2.2.1).
k. When mixers are to shipped mobile (see 5.2.1.2 or 5.2.2.2).

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6.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contract or in accordance with the contract or purchase order requirements.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Definitions.

6.4.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or ashpalt paved highway.

6.4.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.5 Subject (key word) listing.

Aggregate
Concrete
Mixer
Plant
Semitrailer

Custodian:
Navy - YD

Preparing Activity:
Navy - YD

(Project 3895-N238)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-P-28682(YD)	Concrete Plant, Mobile, Semitrailer Mounted, 6 Cubic Meters Bulk Material Capacity (Metric)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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